

2018 JC2 PRELIMINARY EXAMINATIONS

CANDIDATE
NAME

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PDG

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INDEX NUMBER

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H1 BIOLOGY

8876/02

Paper 2 Structured and Free-Response Questions

**11 SEPT 2018
TUESDAY**

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph
Do not use paper clips, highlighters, glue or correction fluid.

Answer all questions in the space provided on the Question Paper.

Section A

Answer **all** questions

Section B

Answer any **one** question

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/13
2	/7
3	/10
4	/11
5	/4
6/7	/15
/60	

Answer **all** questions

- 1 Fig. 1.1 shows the production of all blood cells from Cell X.

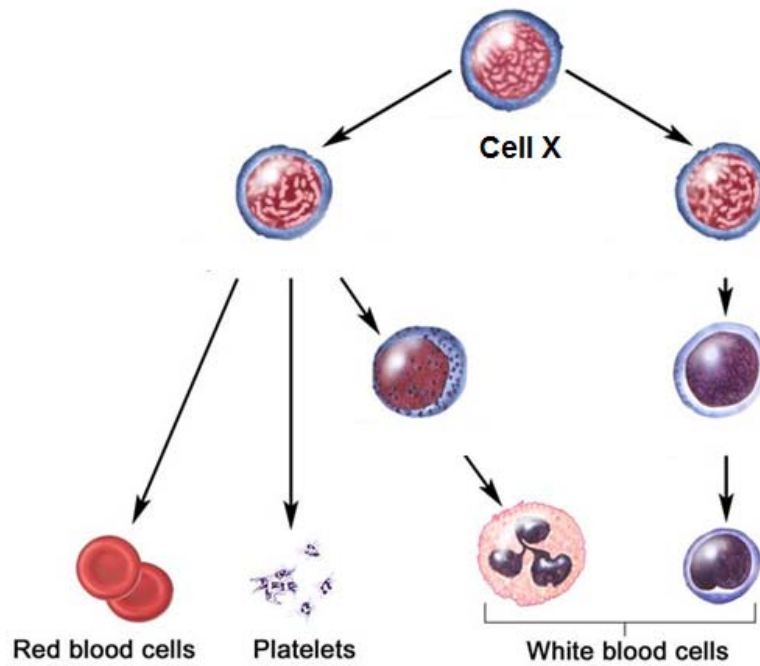


Fig. 1.1

- (a) Identify Cell X and explain why, as shown in Fig. 1.1, it must have the characteristic of a stem cell.

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[4]

- (b) Fig 1.2 is an electron micrograph of a type of white blood cell known as plasma cell, in the process of producing antibodies. Antibodies are proteins secreted by plasma cells.

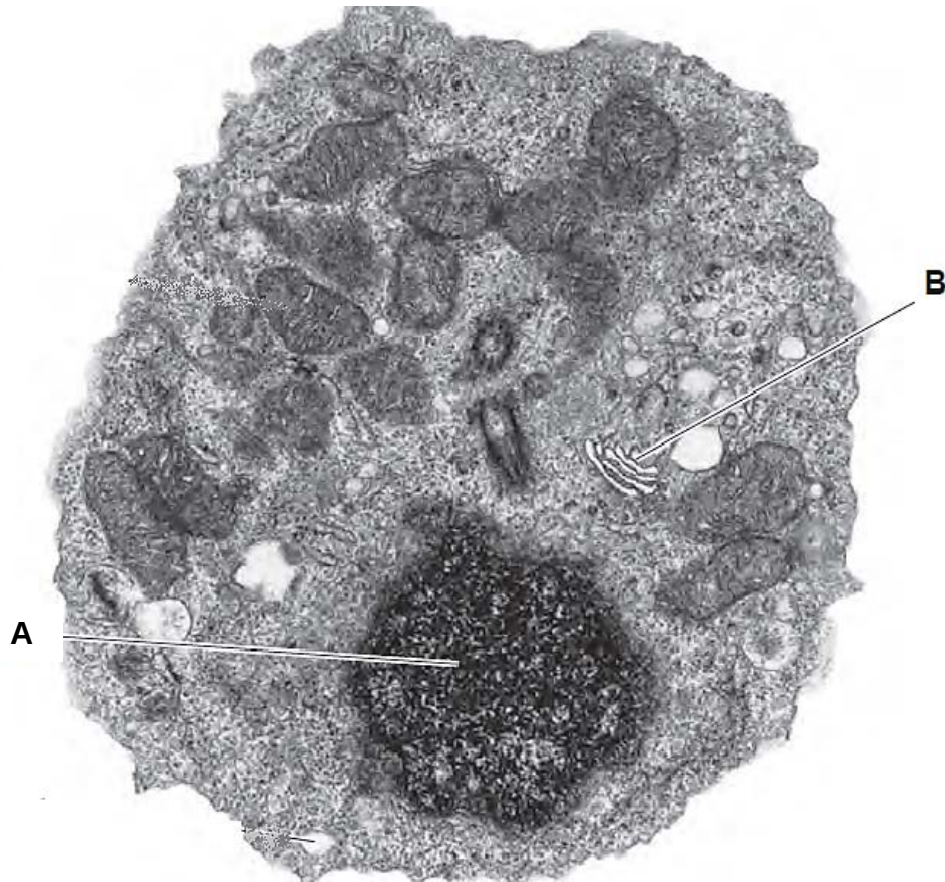


Fig. 1.1

Describe the role of the following organelles in the production of antibodies.

A

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B.....

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(c) Macrophage is another type of white blood cell which is involved in recognising foreign particles and undergoing phagocytosis.

(i) Describe the process of phagocytosis.

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..... [3]

(ii) Suggest how a macrophage recognises specific foreign particles.

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..... [2]

[Total: 13]

2 Fig. 2.1 shows translation occurring on a eukaryotic mRNA.

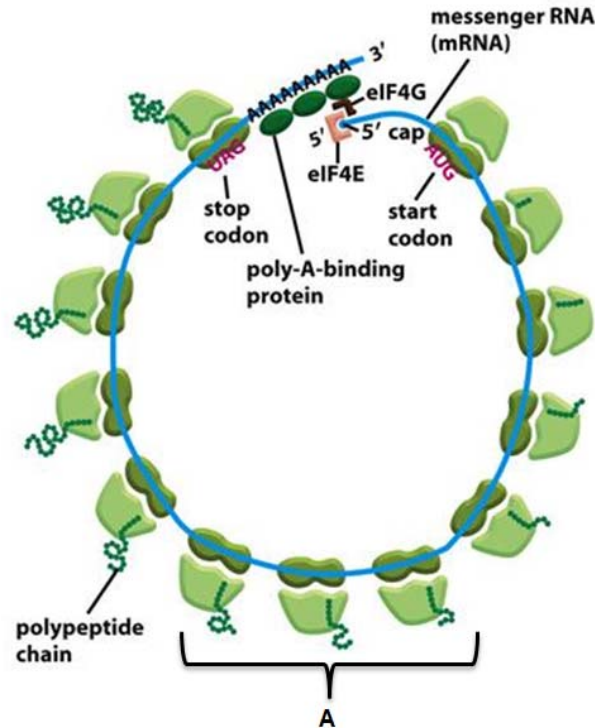


Fig. 2.1

(a) (i) Explain the significance of the pattern of translation, labelled **A** in Fig. 2.1.

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..... [2]

(ii) During initiation of translation, the poly-A tail and 5'cap on the mRNA, together with initiation factors like eIF4E and eIF4G form part of a complex which aid in recruiting ribosomal subunits to mRNA.

With reference to Fig. 2.1, describe the role of poly-A tail and 5' cap in the assembly of ribosomes.

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(b) Describe **three** differences between the process of translation and transcription.

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..... [3]

[Total:7]

3 The ABO blood group system in humans is controlled by a gene with alleles, I^A , I^B and I^O . Each allele lead to production of different antigens (proteins) on the surface of red blood cells.

- Allele I^A , leads to the production of antigen A.
- Allele I^B , leads to the production of antigen B.
- Allele I^O , does not lead to the production of either antigen.

Alleles I^A and I^B are codominant and I^O is recessive to both.

(a) With reference to the ABO blood group, explain what is meant by

(i) codominance.

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..... [2]

(ii) multiple alleles.

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..... [1]

- (b) In humans, a gene that codes for the production of a protein, called factor VIII, is located on the X chromosome. The dominant allele for this gene produces factor VIII, but the recessive allele does not produce factor VIII.

A person who is unable to make factor VIII has haemophilia in which the blood fails to clot properly.

Explain why a man with haemophilia cannot pass haemophilia to his son but may pass haemophilia to his grandson.

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[3]

- (c) A gene for feather colour in chickens is carried on an autosome. This gene has two alleles, black (C^B) and splashed-white (C^W). When a male chicken with black feathers is mated with a female chicken with splashed-white feathers as shown in Fig. 3.1, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.

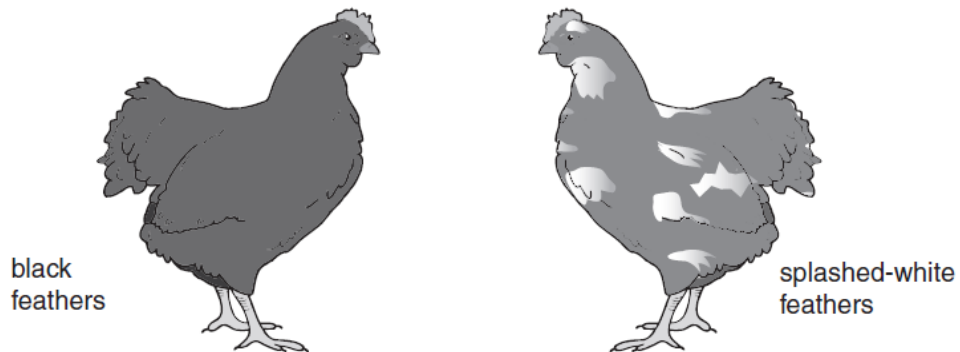


Fig. 3.1

Another gene may cause stripes on feathers (barred feathers), shown in Fig. 3.2. This gene is carried on the X chromosome. The allele for barred feathers (X^A) is dominant to the allele for non-barred feathers (X^a).

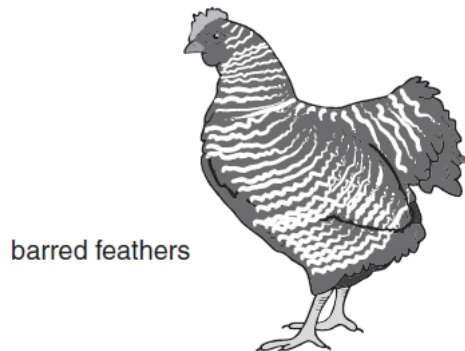


Fig. 3.2

In chickens, the male is homogametic and has two X chromosomes while the female is heterogametic and has one X chromosome and one Y chromosome.

A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

Using the symbols given above, draw a genetic diagram to show this cross.

[4]

[Total: 10]

- 4 To protect crops against insects, many crops such as maize and cotton are genetically modified to contain a gene (Bt) from a bacterium, *Bacillus thuringiensis*. The Bt gene codes for a toxin, which inserts into the insect gut cell membrane, paralyzing the digestive tract and killing the insects which eat the crop. When these genetically modified (GM) crops first became available it was predicted that insect pests would develop resistance to these toxins.

- (a) Explain how Bt resistance may arise and spread in an insect population.

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[4]

- (b) The extent of Bt resistance in insect pest species was surveyed in 2005 and in 2011. The level of resistance in each species was classified according to the highest percentage of resistant individuals recorded in any population anywhere in the world. Three levels of resistance were identified:

- <1%
- 1–6%
- >50%

There were no reports of populations of insect pests having between 6% and 50% of resistant individuals.

The results of the surveys are shown in Table 4.1.

Table 4.1

year	total number of insect pest species surveyed	number of insect pest species susceptible to Bt toxins	number of insect pest species with reported levels of resistance		
			<1%	1–6%	>50%
2005	9	8	0	0	1
2011	13	4	3	1	5

- (i) Discuss whether the data provided can support the hypothesis that rise in Bt resistance is due to introduction of GM crop, including limitations of the data provided.

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[4]

- (ii) Justify the prediction that both insect-caused and non-insect-caused damages to Bt crops will increase with climate change.

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[3]

[Total: 11]

- 5 An analysis of ice cores from the Arctic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 5.1 shows the concentrations of carbon dioxide and methane measured in ice cores, dated between 1000 and 2000 AD.

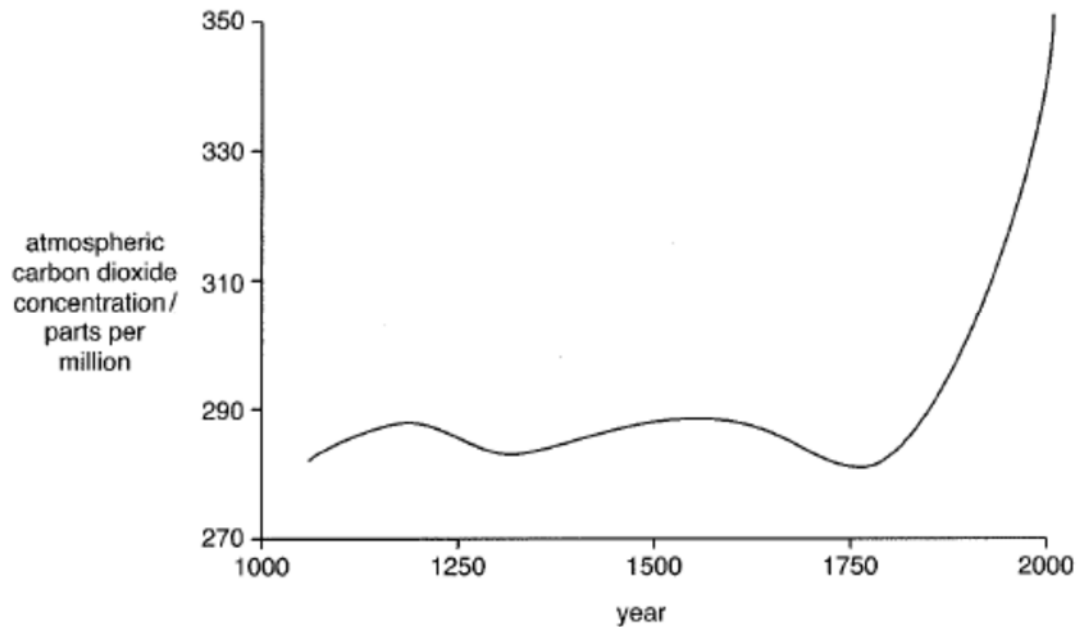


Fig.5.1

- (a) Describe and explain the data in Fig. 5.1 from 1750 AD onwards.

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- (b) Explain how increasing concentrations of gases, such as carbon dioxide and methane, are thought to cause global warming.

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[Total: 4]

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers must be in continuous prose, where appropriate.

6 (a) Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [8]

- A human's genome is constant, whereas the proteome of a human is constantly changing.' Discuss this statement. [7]

7 (a) Discuss the inter-relationships of membranous organelles in eukaryotic cells and suggest the implications of absence of membranous organelles in prokaryotic cells. [8]

- (b)** Explain why more energy is released from the oxidation of triglycerides as compared to the same unit mass of glucose and outline the problems of using triglycerides as the main respiratory substrate instead of glucose in animals. [7]

[Total: 15]

[illegible]

